Stage 1 (Baseline Measurements)	Stage 2 (Microarchitectural Study)	Stage 3 (Performance Framework Development)	Stage 4 (Core Modification & Decoupling)	Stage 5 (Arbitration Module Design)	Stage 6 (System Integration & Evaluation)
Deployment of CVA6 on the Pynq Z2	Analysis of Data and Instruction cache sizes	Automated framework for performance measurement	Development of a modified CVA6 version	Development of the shared resource arbitrator	Evaluation of performance and FPGA utilization (FF/LUT/BRAM)
Baseline SoC Utilization measurements (FF/LUT/BRAM)	Study of Branch Predictor and TLB size influence	Integration of NAS Parallel Benchmarks	Implementation of an external module interface [optional]	Implementation of pipeline stalling logic [optional]	Evaluation of performance NPB and Unixbench
Baseline NPB and Unixbench measurements		Integration of UnixBench suite	Enabling connection for FPU management [optional]	Routing of results between multiple cores [optional]	